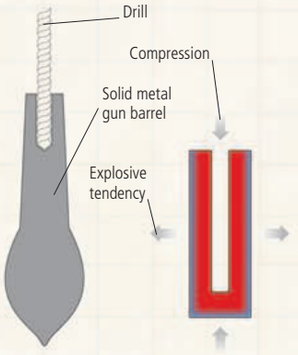
GUN FOUNDING

As the power requirements of cannon increased over the centuries, guns required corresponding strengthening to cope with the greater pressures on the breech. The search for the best cannon-founding method produced two main variants during the 18th and 19th centuries—bore drilling and Rodman casting (here seen applied to a Dahlgren Gun).

BORE DRILLING

The bore-drilling method was pioneered during the mid-18th century, and was a major improvement over the earlier hollow-casting method (casting the gun around a solid core, which was then removed). In bore drilling, the barrel was cast in one solid piece. Then the bore of the gun was drilled out of the metal. The problem with bore drilling was that the gun cooled from the outside in, creating weaknesses in the gun's structure through an outer tension in the metal. This would then place limitations on the power and range of the gun.

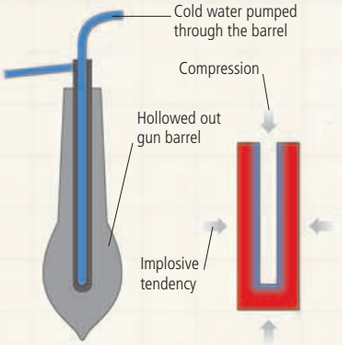


Bore drilling
The bore of the gun is drilled from one piece of solid metal.

Tension
Cooling from outside, compression causes an explosive force.

RODMAN CASTING

Rodman casting was designed by Thomas Jackson Rodman, an American artilleryman. Rodman overcame the problems of bore-drilled guns by casting the cannon over an inner cooling core. This consisted of a cast iron tube, closed at one end but with inflow/outflow valves at the other, a design that allowed the operator to pump a constant flow of cooling water through the tube. The result was that the gun cooled from the inside out, meaning that each successive layer of metal that cooled contracted onto the one beneath it. The subsequent barrel was held together by compression rather than tension, making the gun much stronger and capable of taking heavier charges.



Rodman core
The water-filled core cools the gun from inside out.

Compression
Cooling from inside, compression causes an an implosive force.



EXPLOSIVE SHELL
This explosive shell from the mid-19th century was designed to be fired from a new generation of breech-loaders, in this case a 12-pounder with a rifled barrel.



SHRAPNEL SHOT
This early shrapnel shell featured a hollow cavity filled with small metal balls. The shell would explode in the air and throw out a lethal cloud of anti-personnel shot.



SHRAPNEL SHELL
A mid-19th-century shrapnel shell used in a rifled muzzle-loader. It worked on a time fuse and exploded in flight, creating an airburst effect over an enemy.



HIGH-EXPLOSIVE SHELL
A modern naval shell with a high-explosive filling made from a mixture of TNT and beeswax. The shell's red cone contains dye to help identify which ship had fired the shell.

DAHLGREN BATTERY
Designed by American naval officer John A. Dahlgren, the bulbous breech of the Dahlgren gun allowed shells to be fired at a higher velocity and, consequently, much farther.

